

§3.4 Calibration of the Two-Maturity FTPL Model with Long-End Captive Demand: Methodology

Lucas Leturia

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1 Strategy

The model developed in the preceding sections is sufficiently parsimonious that a calibration to UK steady-state data is both possible and informative. The exercise serves two purposes. First, it disciplines the model by checking whether the augmented FTPL identity holds at empirically plausible parameter values. Second, it quantifies the magnitude of the long-end captive-demand channel relative to the standard surplus channel, producing what becomes the dissertation's central numerical result.

The augmented FTPL identity derived in §3.2 connects six observable quantities — β , δ , V^{liab} , b^L , b^S , s — to a seventh, the long-end price wedge ω , which is not directly observable in market data. The natural calibration strategy is therefore to set the six observables from data and *solve* the identity for the implied wedge. This converts the calibration into a structural-quantification question:

What magnitude of long-end captive-demand wedge is required for the model to rationalize observed UK debt market value at empirically plausible values of all other inputs?

The implied wedge can then be compared to external benchmarks from the term-structure literature on UK long-end yield compression (Greenwood–Hanson–Vayanos and related work) as a check on the model's plausibility. Because UK fiscal and pension-sector data are released annually, the identity can be evaluated period by period to recover a *time series* of implied wedges $\{\omega_t\}_{t=2010}^{2024}$ rather than a single point estimate. The dynamics of this series — its level, persistence, and response to major policy events — constitute the chapter's empirical output.

2 Data Targets and Sources

The six inputs to the identity, and the proposed UK data series for each, are summarized below. Where alternatives exist, both the primary and fallback sources are listed, with a note on the choice.

Real short rate: pinning β

Primary source. Bank of England daily real spot rate curve, 1-year point, derived from index-linked gilts. Available from the BoE's *Yield Curves* statistical release. Take the annual average of daily observations for each calendar year.

Alternative source. 1-year nominal gilt yield (Bloomberg ticker **GUKG1Y** or BoE Statistical Bulletin) minus realized 1-year-ahead RPI or CPI inflation (ONS series **CHAW** for RPI, **D7G7** for

CPI). This gives an ex-post real rate rather than the ex-ante real rate from linkers; the two differ by a small inflation-surprise residual that averages near zero over multi-year windows.

Conversion to β . For each year t , set $\beta_t = 1/(1 + r_t)$ where r_t is the annual real short rate in decimals.

Choice. The linker-based ex-ante measure is preferred because it corresponds to the model's notion of the SDF-implied real short rate. The ex-post measure is used as a robustness check.

Long-bond duration: pinning δ

Primary source. UK Debt Management Office (DMO) publishes the modified Macaulay duration of the gilt portfolio by maturity bucket in its *Quarterly Review* and on its data download portal. Take the duration of the 15+ year segment.

Alternative source. Computed directly from the BoE nominal spot rate curve as the duration of a synthetic 20-year zero-coupon bond, or from Bloomberg gilt portfolio analytics.

Conversion to δ . The model's long bond is a geometric perpetuity. Its Macaulay duration in the deterministic steady state is $D = 1/(1 - \beta\delta)$ years, so the duration target D^{data} implies $\delta_t = (1 - 1/D_t^{\text{data}})/\beta_t$.

Choice. DMO's published duration is the cleanest match to the segment definition (15+ year gilts) the model is intended to represent.

Captive-demand scale: pinning V^{liab}

Primary sources.

- *The Pensions Regulator (TPR), Purple Book*: aggregate UK defined-benefit (DB) pension liabilities on the s179 (Pension Protection Fund) basis. Published annually.
- *Prudential Regulation Authority (PRA) Bank of England Solvency II returns*: aggregate UK insurance technical reserves, with particular attention to the matching-adjustment portfolio of annuity writers.

Alternative source. ONS Pension Funds Statistical Bulletin (MQ5) provides aggregate UK funded occupational pension assets and liabilities at higher frequency, though on a different valuation basis.

Conversion to $V^{\text{liab}}/\bar{y}$. Sum aggregate DB pension liabilities and insurance technical reserves; divide by nominal GDP (ONS series YBHA).

Choice. The Purple Book is the standard regulatory measure and is what the LDI mandate framework is benchmarked against. The s179 basis is preferred over the buyout basis because it corresponds to the regulatory liability that drives LDI demand, not the market-clearing buyout cost.

Long and short gilt market values: pinning b^L , b^S , χ

Primary source. DMO publishes daily and quarterly statistics on the outstanding gilt portfolio by maturity bucket, including both face value and (separately) market value. Available from www.dmo.gov.uk.

Alternative source. Bloomberg's UK gilt market value series, aggregated by maturity bucket. Or BoE's Markets and Banking statistics for the consolidated public-sector gilt portfolio.

Conversion. Compute long-gilt market value / nominal GDP for $Q^L b^L / \bar{y}$, and short and medium gilt market value / nominal GDP for $Q^S b^S / \bar{y}$. The long-share χ is the ratio of long-gilt to total gilt market value.

Choice. Market value, not face value, is the correct empirical analog of the model's b^L once the unit convention is fixed (see the note on units in §4 below). The DMO market-value series is the appropriate primary source. Excludes index-linked gilts in the headline calibration; sensitivity exercise includes them.

Primary balance: pinning s

Primary source. Office for Budget Responsibility (OBR), *Public Sector Finances Databank*, primary balance series. Available from OBR website at annual frequency.

Alternative source. ONS public sector finances release (PSF), which provides cyclically-adjusted and headline primary balance measures.

Conversion. Take primary balance as percentage of nominal GDP; divide by 100.

Choice. OBR series is preferred because it is the same fiscal accounting used in the OBR's Economic and Fiscal Outlook publications that anchor the empirical chapter (Ch. 4).

Nominal GDP: denominator for ratios

Primary source. ONS, gross domestic product at current prices, annual (series YBHA).

Alternative source. IMF World Economic Outlook UK series; OECD.

Choice. ONS is the official source and is what the OBR and DMO use as the denominator in their published ratios.

3 Sample and Frequency

The sample is calendar years 2010–2024, fifteen observations. This window is chosen for three reasons:

1. 2010 is the year of OBR's creation, providing a consistent fiscal-forecast benchmark from this date forward (relevant for cross-reference with Ch. 4).
2. The window covers a complete monetary-policy cycle (ZLB period, post-pandemic normalization, 2022 inflation shock) without spanning major regulatory regimes (Basel III phase-in is contained within the window).
3. Fifteen annual observations are sufficient to characterize the dynamics of the implied wedge series while remaining tractable for the steady-state framework.

Annual frequency is chosen over quarterly because the most important inputs (Purple Book, PRA returns, OBR fiscal data) are released annually and would require extensive interpolation to map to quarterly. The model's steady-state structure does not require dynamics, so the loss of frequency is not material for this exercise.

4 Unit Convention

A note on units is necessary before the algorithm. The model's long bond is a geometric perpetuity paying nominal coupons $1, \delta, \delta^2, \dots$ starting at $t + 1$. The fundamental steady-state

price of one such bond is

$$Q^{L,\text{fund}} = \frac{\beta}{1 - \beta\delta},$$

which evaluates to approximately 19.8 at $\beta = 0.99$, $\delta = 0.96$. This is far above unity because one “unit” of the model long bond is a perpetuity capturing two decades of discounted future coupons — it is not directly comparable to one pound of face-value gilt.

The units-invariant object that maps cleanly to data is the *market value* $Q^L b^L$, which can be measured directly as the GDP-normalized market value of long-dated UK gilts. Given Q^L (computed from β , δ , and ω), the stock b^L in model fresh-bond-equivalent units is recovered as $b^L = (Q^L b^L)^{\text{data}} / Q^L$.

Throughout the calibration, “market value of long debt” refers to the GDP-normalized observable $Q^L b^L / \bar{y}$, and the model stock b^L is treated as an internally-computed model object derived from this observable and the implied price.

5 Pseudocode

The calibration loops over years $t = 2010, \dots, 2024$ and applies a fixed algebraic operation at each t . There is no optimization, no iteration over parameters, no estimation in the statistical sense — only a closed-form solve for the implied wedge given the year’s data. The pseudocode is therefore short.

Listing 1: Calibration pseudocode

```
# Inputs (compiled from sources in Section 2):
# real_short_rate[t]      annual real 1y rate, in decimals
# long_duration[t]        long-gilt segment duration, in years
# V_liab[t]               (pension + insurer liabilities) / GDP
# long_gilt_mkt_val[t]    long-gilt market value / GDP
# short_gilt_mkt_val[t]   short+medium-gilt market value / GDP
# primary_balance[t]      primary balance / GDP, in decimals
# pi_bar                  steady-state gross inflation (= 1)

# Output:
# omega[t]                implied long-end wedge for year t
# R[t]                    implied rent flow / GDP for year t
# Q_L[t]                  implied long-bond price for year t
# Q_L_fund[t]             fundamental long-bond price for year t
# lambda_L[t]             implied mandate intensity for year t
# zeta[t]                 long-end repression share for year t
# v[t]                    real debt market value / GDP for year t

for t in range(2010, 2025):

    # Step 1: deep parameters from data
    beta[t] = 1.0 / (1.0 + real_short_rate[t])
    delta[t] = (1.0 - 1.0/long_duration[t]) / beta[t]

    # Step 2: fundamental long-bond price
    Q_L_fund[t] = beta[t] / (1.0 - beta[t] * delta[t])

    # Step 3: real debt market value (start-of-period, using
    # data-observed market values and steady-state pricing)
    v[t] = short_gilt_mkt_val[t] + long_gilt_mkt_val[t]
    # In steady state with pi=1, the start-of-period real
    # market value equals the end-of-period market value
    # carried forward. Equivalent up to inflation correction
    # for the data-mapping; see appendix.
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# Step 4: solve augmented FTPL for implied rent
R[t] = (1.0 - beta[t]) * v[t] - primary_balance[t]

# Step 5: implied wedge from rent and market-clearing
# R = (omega / (1+omega)) * lambda_L * V_liab
# and lambda_L * V_liab = Q_L * b_L = long_gilt_mkt_val
rent_share = R[t] / long_gilt_mkt_val[t]
omega[t] = rent_share / (1.0 - rent_share)

# Step 6: derived quantities
Q_L[t] = (1.0 + omega[t]) * Q_L_fund[t]
lambda_L[t] = long_gilt_mkt_val[t] / V_liab[t]
b_L[t] = long_gilt_mkt_val[t] / Q_L[t]
yield_wedge[t] = (1.0/Q_L_fund[t] - 1.0/Q_L[t]) # bps in level
zeta[t] = (R[t] / (1.0 - beta[t])) / v[t]

# Sanity checks (asserted after the loop):
# omega[t] > 0 for all t (Assumption 1 binds)
# lambda_L[t] in (0, 1) for all t
# beta[t] * delta[t] < 1 for all t (long-bond price finite)

# Output: tables of {omega, R, lambda_L, yield_wedge, zeta} by year.

```

The structure is exactly as expected: one equation per period, no iteration, no optimization. The output is a time series of seven derived quantities $\{Q_L, Q_L^{\text{fund}}, \omega, R, \lambda^L, \text{yield wedge}, \zeta\}$ for each calendar year, plus the time series of the two deep parameters $\{\beta, \delta\}$.

6 What the Output Looks Like

The headline output is the time series of ω_t and the associated yield wedge. The dissertation will report this as a single time-series plot with annotations for major policy events:

- *Pre-2016*: ZLB period with low real rates; expected to show modest implied wedge consistent with low fiscal-capacity gap.
- *2016–2019*: post-Brexit fiscal loosening; expected to show widening wedge as primary balance deteriorates.
- *2020–2021*: COVID-driven debt expansion with continued QE; expected to show further wedge increase.
- *September 2022*: mini-budget episode; the implied wedge should spike if pension-sector data is timely enough to capture LDI stress (likely visible in 2022 annual data).
- *2023–2024*: normalization following BoE intervention and fiscal reversal.

Tables will report:

1. Calibration inputs by year (six data series).
2. Derived quantities by year (ω , R , λ^L , yield wedge, ζ).
3. Comparative-statics counterfactuals at the 2010–2024 average: β perturbations, s perturbations, V^{liab} perturbations.

7 Sanity Checks and Limitations

The calibration imposes one substantive economic assumption: that $\omega_t > 0$ in every period (the mandate binds, Assumption 1 of the model). The algorithm reports ω_t for each year regardless of sign; if any year delivers $\omega_t \leq 0$, this indicates that the augmented FTPL identity would hold at that year's data even without the captive-demand channel — a finding of independent interest, and a signal that the model's binding-mandate assumption is empirically nontrivial.

Three caveats merit explicit statement:

1. The calibration assumes steady-state pricing each period. Annual frequency makes this a reasonable approximation but not an exact one: within-year fluctuations in gilt yields, pension liabilities, and the primary balance are smoothed out.
2. The augmented FTPL identity is taken as a within-period restriction. A fully dynamic calibration would impose the no-Ponzi terminal condition forward-looking; this is the subject of the estimation strategy in §3.5.
3. σ is not identified by the steady-state data and is set to unity by convention. This parameter becomes identified only when dynamic moments are introduced.

8 Connection to Estimation (§3.5)

The calibration delivers a series $\{\omega_t\}$ derived from the steady-state identity. The estimation exercise outlined in §3.5 instead treats the model as a dynamic stochastic system, log-linearizes around steady state, and uses the Kalman-filter likelihood to recover the structural parameters jointly. The calibration in this section provides:

1. A consistency check on the deep parameters β and δ that will enter the estimation.
2. A prior over the steady-state values that anchor the log-linearization.
3. A descriptive object — the time series of implied wedges — against which the estimated structural shocks can be compared.

The two exercises are complements rather than substitutes. The calibration is descriptive and identity-based; the estimation is statistical and model-based. Both are required for the dissertation to make a quantitative contribution.